PRIYANSHU MISHRA

AI Engineer | Full-Stack | BSc Computer Science, UNSW Sydney
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PROFESSIONAL SUMMARY

Computer Science graduate from UNSW with Distinction and experience building AI, full-stack and machine learning systems. At ArmsOA, built Claude-powered agentic workflows with live-state context, evidence-grounded outputs and human-in-the-loop controls, integrated into a Python/FastAPI, React/TypeScript and Docker network resilience platform backed by 500+ regression tests. Previously developed an end-to-end ML pipeline for encrypted network traffic analysis achieving 92% accuracy. Strong across AI/LLM engineering, backend systems, full-stack development and reliable, auditable AI applications.

EDUCATION

Bachelor of Science (Computer Science) · UNSW Sydney · Distinction WAM (78) *Sep 2023 – Aug 2026*
Coursework: Machine Learning, Computer Vision, Software Engineering, Computer Networks, Security Engineering

TECHNICAL SKILLS

- **AI and LLM systems:** agentic workflows, context injection from live state, human-in-the-loop approval, confidence scoring, model evaluation, deterministic engines alongside LLM agents
- **Machine learning:** PyTorch, scikit-learn, ensemble methods, feature engineering, imbalanced classification, U-Net (ResNet50), SegFormer, OpenCV
- **Backend and data:** Python, SQL, FastAPI, OpenAPI 3.0, JSON Schema contracts, WebSockets, Pandas, NumPy, Docker, Git
- **Frontend:** TypeScript, JavaScript, React, Next
- **Networks:** Scapy, pcapng analysis, WebExtensions API, Selenium

RELEVANT EXPERIENCE

Software Engineer · ArmsOA, Canberra *Jun 2026 – Sep 2026*

- Built a real-time network resilience simulator (Python/FastAPI, React, TypeScript, Docker) for emergency response operators, modelling the network as a graph of nodes, multi-link edges and mission-prioritised bearers across 4G/5G, microwave, SATCOM and UHF/VHF under failure and environmental stress.
- Owned the AI interface: Tactical Analytics and Recovery COA agents on Claude Opus/Sonnet with live-state context injection, evidence-grounded outputs and persistent sessions. Agents recommend; operators approve. No recovery action runs without a human decision.
- Engineered deterministic PACE + CPDS drone recovery, multi-run COA simulation, candidate-specific confidence scoring, configurable fleet inventory and temporary bearer restoration while preserving auditable physical-link failures.
- Built effective-recovery observability across Monitor/Report, mission analytics and exports; translated client specifications into production APIs, operator controls and 500+ backend regression tests.

AI Researcher · UNSW School of Electrical Engineering, Sydney *Jun 2025 – Feb 2026*

- Built an end-to-end pipeline detecting DNS protocol downgrades in live encrypted traffic: pcapng ingestion, feature engineering and an ML classifier separating DoH/DoT/DoQ from plaintext fallback, reaching 92% accuracy on real captures.
- Linked encrypted traffic back to browsing sessions using a Firefox WebExtension, a Python/Scapy sniffer and a Selenium harness, attaching a confidence score to each correlation instead of a hard label.
- Owned the full model lifecycle (scoping, data pipeline, architecture, evaluation, iteration) and presented findings to both engineers and non-technical infrastructure stakeholders.

Resident Advisor · Shalom College, UNSW, Sydney *Feb 2026 – Present*

- First point of contact for 130 residents: on-call duty, welfare checks and confidential incident escalation.

PROJECTS

Insurance Claims Fraud Detection | Atlassian × Allianz Datathon 2024 *Highest accuracy, Finished in top 5*

- Built a scikit-learn ensemble classifier on a heavily imbalanced real claims dataset, tuned to keep false positives low because each one carried direct cost to a genuine customer; 97% accuracy. Presented model rationale and risk trade-offs to Atlassian and Allianz engineers.

Wheat Crop Segmentation Benchmark | Python, PyTorch, OpenCV

- Compared SLIC, Random Forest, U-Net/ResNet50 and SegFormer-B0 on accuracy, latency and compute cost; best model reached 0.91 IoU on held-out test data.

Quant AI Live Intelligence | Python, FastAPI, TypeScript, Redis, Docker

- Turns a company name into a source-attributed forecast from price history, market news and Reddit via documented APIs. Every response carries provenance, validation metrics and explicit provider degradation instead of fabricated data

ACHIEVEMENTS

- UNSW Taste of Research Scholarship (competitive grant) to lead original research in AI agents for network observability
- Top 1% nationally in STEM and English Olympiads; State Topper in Science (IAIS, UNSW); High Distinction, Australian National Chemistry Quiz (RACI)